NO.17 COOLING SYSTEM CONCERNS-OVERHEATING [SKYACTIV-G 1.3, SKYACTIV-G 1.5]

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17	COOLING SYSTEM CONCERNS-OVERHEATING
DESCRIPTION	• Engine runs at higher than normal temperature/overheats.
POSSIBLE CAUSE	• Engine heat generation amount is excessive — Engine speed is high and remains high for continuous long periods • PCM DTC is stored • Improper operation of A/C system • Insufficient air blow volume — Cooling fan inoperative — Fan rotation speed decreases (such as generator malfunction) due to battery voltage decrease • Cooling system cap malfunction • Cooling fan motor assembly malfunction • Cooling fan motor connector malfunction • Improper indication of low engine coolant temperature indicator light and/or high engine coolant temperature warning light — ECT sensor malfunction — Instrument cluster malfunction • Blown fuse • Engine coolant system malfunction — Improper engine coolant level — Engine coolant performance deterioration — Coolant leakage (engine internal, external) — Improper water/anti-freeze mixture — Thermostat malfunction (not fully open) — Radiator malfunction (clogging) • Leakage around heater unit in passenger compartment • Drive belt damage

Diagnostic Procedure

STEP	INSPECTION	RESULTS	ACTION
1	**INSPECT VEHICLE CONDITION FOR EFFECT ON MALFUNCTION** • Verify how the customer drives the vehicle by asking the customer the following: — Engine speed is high and remains high for continuous long periods like when climbing a long, steep grade. • Is the engine generating excessive heat when driven?	Yes	Explain to the customer that the vehicle is normal and give them advice how to use vehicle and provide a specific example of the conditions in which the engine overheats.
		No	Go to the next step.
2	**VERIFY PCM DTC** • Retrieve any DTCs using the M-MDS. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
		No	Go to the next step.
3	**DETERMINE IF MALFUNCTION CAUSE IS A/C SYSTEM OR OTHER** • Start the engine and run it at idle speed. • Turn the A/C switch off. • Does the A/C compressor disengaged?	Yes	Go to the next step.
		No	Perform the symptom troubleshooting “NO.24 A/C IS ALWAYS ON OR A/C COMPRESSOR RUNS CONTINUOUSLY”. (See NO.24 A/C IS ALWAYS ON OR A/C COMPRESSOR RUNS CONTINUOUSLY [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)

STEP	INSPECTION	RESULTS	ACTION
4	**INSPECT ECT SENSOR SIGNAL** • Access the ECT PID using the M-MDS. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Compare the ECT PID and low engine coolant temperature indicator light and high engine coolant temperature warning light operation. • Is the ECT PID value same as indicator/warning light readings?	Yes	Go to the next step.
		No	Inspect the ECT sensor. (See ENGINE COOLANT TEMPERATURE (ECT) SENSOR INSPECTION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • If there is any malfunction: — Replace the ECT sensor. (See ENGINE COOLANT TEMPERATURE (ECT) SENSOR REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • If there is no malfunction: — Perform the instrument cluster symptom troubleshooting “LOW ENGINE COOLANT TEMPERATURE INDICATOR LIGHT/HIGH ENGINE COOLANT TEMPERATURE WARNING LIGHT ILLUMINATES OR FLASHES CONTINUOUSLY”. (See LOW ENGINE COOLANT TEMPERATURE INDICATOR LIGHT/HIGH ENGINE COOLANT TEMPERATURE WARNING LIGHT ILLUMINATES OR FLASHES CONTINUOUSLY [INSTRUMENT CLUSTER].)
5	**INSPECT COOLING FAN CONTROL SYSTEM OPERATION** • Perform the Cooling Fan Control System Inspection. (See ENGINE CONTROL SYSTEM OPERATION INSPECTION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Does the cooling fan control system operate properly?	Yes	Go to the next step.
		No	Repair or replace the malfunctioning part according to the inspection results.
6	**INSPECT COOLING SYSTEM CAP** • Inspect the cooling system cap. (See COOLING SYSTEM CAP INSPECTION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Is there any malfunction?	Yes	Replace the cooling system cap.
		No	Go to the next step.
7	**INSPECT RELATED PART CONDITION** • Inspect the following: — Engine coolant level — Engine coolant performance — Coolant leakage (engine internal, external) — Water and anti-freeze mixture — Thermostat (not fully open) — Radiator (clogging) — Fuses • Is there any malfunction?	Yes	Service if necessary. • Repeat this step.
		No	Go to the next step.
8	**VERIFY IF MALFUNCTION CAUSED BY LACK OF ENGINE COOLANT** • Inspect the engine coolant level. (See ENGINE COOLANT LEVEL INSPECTION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Is there any malfunction?	Yes	Add engine coolant and verify that there is no engine coolant leakage. (See ENGINE COOLANT REPLACEMENT [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See ENGINE COOLANT LEAKAGE INSPECTION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • If there is any malfunction: — Repair or replace the malfunctioning part according to the inspection results.
		No	Go to the next step.

STEP	INSPECTION	RESULTS	ACTION
9	**DETERMINE IF MALFUNCTION CAUSE IS THERMOSTAT OR OTHER** • Verify the radiator hose tension. **Warning** **• To prevent burns, use a cloth with your hand to verify the tension of the radiator hose.** • After the engine warms up, does the engine coolant circulate to the radiator hose?	Yes	Go to Step 11.
		No	Go to the next step.
10	**INSPECT THERMOSTAT** • Inspect the thermostat. (See THERMOSTAT INSPECTION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Is the thermostat normal?	Yes	Inspect the water pump. • If there is any malfunction: — Repair or replace the malfunctioning part according to the inspection results. (See WATER PUMP REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
		No	Replace the thermostat. (See THERMOSTAT REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
11	**INSPECT DRIVE BELT** • Inspect the drive belt. (See DRIVE BELT INSPECTION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Is the indicator mark on the drive belt auto tensioner within the normal range?	Yes	Go to the next step.
		No	Replace the drive belt. (See DRIVE BELT REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
12	**INSPECT RADIATOR FOR CLOGGING** • Inspect for clogging in the radiator. • Is there any malfunction?	Yes	Repair or replace the malfunctioning part according to the inspection results.
		No	Replace the engine coolant. (See ENGINE COOLANT REPLACEMENT [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
13	Verify the test results. • If normal, return to the diagnostic index to service any additional symptoms. (See SYMPTOM DIAGNOSTIC INDEX [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • If a malfunction remains, inspect the related Service Information and perform the repair or diagnosis. — If the vehicle is repaired, troubleshooting is completed. — If the vehicle is not repaired or additional diagnostic information is not available, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)		